

# Scalable Transformer Architectures for Genomic Sequence Analysis

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## Abstract

Biological sequence modeling has seen rapid advancements through large self-supervised models. However, existing architectures struggle with quadratic complexity when analyzing long genomic contigs. In this paper, we propose HelixFormer, a sub-quadratic attention mechanism optimized for chromatin structure. We evaluate HelixFormer on variant effect prediction and promoter accessibility across 54 human tissues. Our benchmarks demonstrate a 2.4x speedup and 18% improvement on rare variant pathogenicity classification.

## 1. Introduction and Background

Understanding the non-coding genome remains one of the fundamental challenges in modern computational genetics. Over 98% of human genomic variants fall into non-coding regions, complicating clinical interpretation. Deep learning approaches have mapped regulatory syntax directly from nucleotide sequences. Despite these successes, standard dense transformers fail on sequences extending beyond 32 kilobases. Recent works (Smith et al., 2023) explored state-space models, but suffered degradation on splice site junctions.

## 2. Methods and Objective Function

Let  $X = (x_1, \dots, x_N)$  denote a nucleotide sequence where each  $x_i$  in  $\{A, C, G, T, N\}$ .

The multi-scale attention operator computes contextual embeddings over structured intervals.

We parameterize the loss function with tissue-specific weighted cross entropy:

$$L_{\text{total}} = \alpha * L_{\text{expression}} + (1 - \alpha) * L_{\text{chromatin}}$$

## 3. Empirical Results

Table 1 highlights our performance against baseline foundation models across ClinVar benchmark sets.

HelixFormer achieves an AUROC of 0.892 on pathogenic non-coding SNVs, surpassing previous state-of-the-art.

Ablation experiments confirm that rotational positional embeddings are critical for splice donor sites.